

CHAPTER SIX

REGRESSION AND CORRELATION

6.1. Simple Linear Regression

Regression may be defined as *the estimation or prediction of the unknown value of one variable from the known values of one or more variables*. The variable whose values are to be estimated or predicted is known as *dependent* or *explained variable* while the variables which are used in determining the value of the dependent variable are called *independent* or *predictor variables*.

The regression study that involves only two variables is called simple regression and the regression analysis that studies more than two variables is called multiple regressions. If the relationship between the two variables can be described by a straight line then the regression is known as *linear regression* otherwise it is called non-linear.

- *The regression analysis involving only two variables and having a linear relationship is called Simple Linear Regression. This linear relationship between the two variables is represented by a straight line.*

Regression Line (Line of Regression): is the line that gives the best estimate of one variable for any given value of another variable. The regression line which is used to predict the values of Y for any given value of X is called regression line of Y on X. Similarly the regression line which is used to predict the values of X for any given value of Y is called regression line of X on Y.

Regression Equation: is a mathematical equation that defines the relationship between two variables.
Regression of Y on X

$$\text{Model: } Y = \alpha + \beta X + \epsilon$$

Where Y is the dependent variable

X is the independent variable

α is constant term(intercept)

β is slope(change in Y for a unit change in X)

ϵ is the error term

To estimate the regression coefficients (α and β), the procedure is minimizing the sum of the squares of the errors. Let the estimated model be $\hat{Y} = a + bX$. Then, from sample data the values of a (estimate of α) and b (estimate of β) can be obtained as follows:

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} \quad \text{and} \quad a = \bar{Y} - b \bar{X}.$$

Interpretation of the slope (b)

1. If b is positive, there is a direct relationship between the two variables.
2. If b is zero, there is no linear relationship between the two variables.
3. If b is negative, there is an indirect relationship between the two variables.

6.1.1. Curve Fitting

Curve fitting, also known as regression analysis, is used to find the "best fit" line or curve for a series of data points. Most of the time, the curve fit will produce an equation that can be used to find points

anywhere along the curve. In some cases, you may not be concerned about finding an equation. Instead, you may just want to use a curve fit to smooth the data and improve the appearance of your plot. KaleidaGraph provides curve fits that can be used in both of these scenarios.

The curve fits included in KaleidaGraph can be divided into three main categories: Least Squares curve fits, nonlinear curve fits, and smoothing curve fits. This section provides an overview of each category.

Least Squares is a method of curve fitting that has been popular for a long time. Least Squares minimizes the square of the error between the original data and the values predicted by the equation. While this technique may not be the most statistically robust method of fitting a function to a data set, it has the advantage of being relatively simple (in terms of required computing power) and of being well understood. The major weakness of the Least Squared method is its sensitivity to outliers in the data. If a data point is widely different from the majority of the data, it can skew the results of the regression. For this reason, the data should always be examined for reasonableness before fitting. KaleidaGraph's Data Selection tool provides a simple method of graphically removing outliers from a plot.

The five Least Squares fits available in KaleidaGraph are: Linear, Polynomial, Exponential, Logarithmic, and Power. Section 6.1.2 contains a description of linear fit.

6.1.2. Least Squares Method

The **least squares method** is a procedure for using sample data to find the estimated regression equation.

Illustration: Suppose data were collected from a sample of 10 Armand's Pizza Parlor restaurants located near college campuses. For the i^{th} observation or restaurant in the sample, x_i is the size of the student population (in thousands) and y_i is the quarterly sales (in thousands of dollars). The values of x_i and y_i for the 10 restaurants in the sample are summarized in Table 6.1.

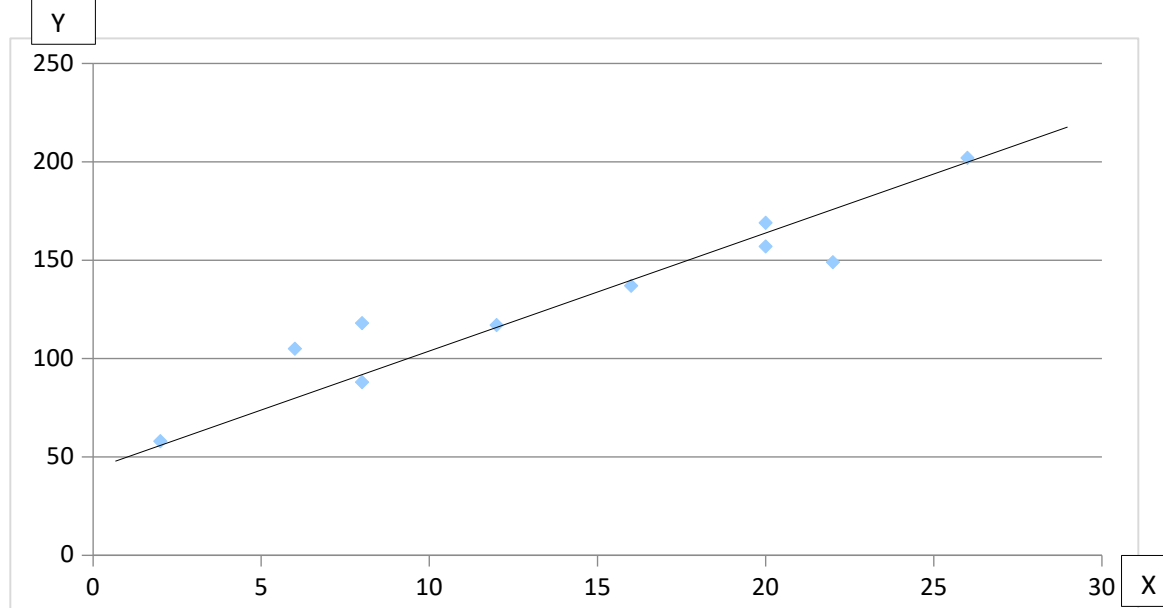
Restaurant (i)	Student Population (1000s) x_i	Quarterly Sales (\$1000s) y_i
1	2	58
2	6	105
3	8	88
4	8	118
5	12	117
6	16	137
7	20	157
8	20	169
9	22	149
10	26	202

Solution:

Figure 631 is a scatter diagram of the data in Table 6.1. Student population is shown on the horizontal axis and quarterly sales are shown on the vertical axis. **Scatter diagrams** for regression analysis are constructed with the independent variable x on the horizontal axis and the dependent variable y on the vertical axis. The scatter diagram enables us to observe the data graphically and to draw preliminary conclusions about the possible relationship between the variables.

What preliminary conclusions can be drawn from Figure 6.1? Quarterly sales appear to be higher at campuses with larger student populations. In addition, for these data the relationship between the size of the student population and quarterly sales appears to be approximated by a straight line; indeed, a positive linear relationship is indicated between x and y .

Figure 6.1. Scatter diagram of student population and quarterly sales for Armand's pizza parlors



We therefore choose the simple linear regression model to represent the relationship between quarterly sales and student population. Given that choice, our next task is to use the sample data in Table 6.1 to determine the values of a and b in the estimated simple linear regression equation. For the i^{th} restaurant, the estimated regression equation provides:

$$\hat{Y} = a + bX.$$

Restaurant (i)	Student Population (1000s) x_i	Quarterly Sales (\$1000s) y_i	$x_i * y_i$	x_i^2
1	2	58	116	4
2	6	105	630	36
3	8	88	704	64
4	8	118	944	64
5	12	117	1404	144
6	16	137	2192	256
7	20	157	3140	400
8	20	169	3380	400
9	22	149	3278	484
10	26	202	5252	676
	$\sum x_i = 140$	$\sum y_i = 1300$	$\sum x_i * y_i = 21040$	$\sum x_i^2 = 2528$

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} \quad \text{and}$$

$$a = \bar{Y} - b \bar{X}$$

$$b = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}$$

$$= \frac{10(21040) - (140 \cdot 1300)}{10(2528) - (140)^2}$$

$$= 5$$

$$a = \bar{Y} - b \bar{X}$$

$$= \frac{\sum y_i}{n}$$

$$= \frac{1300}{10} = 130$$

$$= \frac{\sum x_i}{n}$$

$$= \frac{140}{10} = 14$$

$$a = \bar{Y} - b \bar{X}$$

$$a = 130 - 5(14) = 60$$

$$\hat{Y} = a + bX$$

$$\hat{Y} = 60 + 5X$$

Hence, we can predict quarterly sales for a restaurant near to the restaurant using this linear equation. For example, if we wanted to predict quarterly sales for a restaurant to be located near a campus with 16,000 students, we would compute:

$$\hat{Y} = 60 + 5(16) = 140$$

Hence, we would predict quarterly sales of \$140,000 for this restaurant.

6.2. Linear Correlation

6.2.1. Correlation Coefficient

Whenever a correlation coefficient is calculated, the data should be plotted on a scatter plot. No single numerical measure can substitute for the visual insight gained from a plot. Many different patterns can

produce the same correlation coefficient, and similar strengths of relationships can produce differing coefficients, depending on the curvature of the relationship.

Characteristics of Correlation Coefficients

Correlation coefficients measure of the strength of association between two continuous variables. Of interest is whether one variable generally increases as the second increases, whether it decreases as the second increases, or whether their patterns of variation are totally unrelated. Correlation measures observed co-variation.

Measures of correlation (here designated in general as ρ) have the characteristic of being dimensionless and scaled to lie in the range $-1 \leq \rho \leq 1$. When there is no correlation between two variables, $\rho = 0$.

When one variable increases as the second increases, ρ is positive. When they vary in opposite directions, ρ is negative. The significance of the correlation is evaluated using a hypothesis test:

$$H_0: \rho = 0 \quad \text{versus} \quad H_a: \rho \neq 0$$

When one variable is a measure of time or location, correlation becomes a test for temporal or spatial trend.

Correlation coefficients (denoted r) are statistics that quantify the relation between X and Y in unit-free terms. When all points of a scatter plot fall directly on a line with an upward incline, $r = +1$; When all points fall directly on a downward incline, $r = -1$.

Such perfect correlation is seldom encountered. We still need to measure *correlation strength*, –defined as the *degree* to which data point adhere to an imaginary trend line passing through the “scatter cloud.” Strong correlations are associated with scatter clouds that adhere closely to the imaginary trend line. Weak correlations are associated with scatter clouds that adhere marginally to the trend line.

The correlation coefficient is a measure of the strength and the direction of a linear relationship between two variables. The symbol r represents the sample correlation coefficient. A formula for r is:

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \quad \text{Sample correlation coefficient}$$

where n is the number of pairs of data.

The population correlation coefficient is represented by ρ (the lowercase Greek letter rho, pronounced “row”).

The range of the correlation coefficient is -1 to 1, inclusive. When x and y have a strong positive linear correlation, r is close to 1. When x and y have a strong negative linear correlation, r is close to -1. When x and y have perfect positive linear correlation or perfect negative linear correlation, r is equal to 1 or -1, respectively.

When there is no linear correlation, r is close to 0. It is important to remember that when r is close to 0, it does not mean that there is no relation between x and y , just that there is no linear relation.

Correlation coefficient can be interpreted as:

Degree	Positive	Negative
Perfect	+1	-1
Very high	+0.75 to +1	-0.75 to -1
High	+0.5 to + 0.75	- 0.5 to -0.75
Low	+0.25 to + 0.5	-0.25 to – 0.5
Very low	0 to +0.25	0 to -0.25
Absent	0	0

Example: A director of alumni affairs at a small college wants to determine whether there is a linear relationship between the number of years alumni have been out of school and their annual contributions (in thousands of dollars). The data are shown in the table below.

Number of years out of school, x	1	10	5	15	3	24	30
Annual contribution (in 1000s of \$), y	12.5	8.7	14.6	5.2	9.9	3.1	2.7

Required:

Calculate the correlation coefficient for the number of years out of school and annual contribution data.

Interpret the result in the context of the data.

A. Identify n and use a table to calculate Σx , Σy , Σxy , Σx^2 , and Σy^2

B. Use the resulting sums and n to calculate r.

C. Interpret the result in the context of the data.

Number of years out of school, x	Annual contribution (in 1000s of \$), y	Σxy	X^2	Y^2
1	12.5	12.5	1	156.25
10	8.7	87	100	75.69
5	14.6	73	25	213.16
15	5.2	78	225	27.04
3	9.9	29.7	9	98.01
24	3.1	74.4	576	9.61
30	2.7	81	900	7.29
		435.6	1,836	587.05

a. $\Sigma x = 1 + 10 + 5 + 15 + 3 + 24 + 30 = 88$

$$\Sigma y = 12.5 + 8.7 + 14.6 + 5.2 + 9.9 + 3.1 + 2.7 = 56.7$$

$$\Sigma xy = 435.6, \quad \Sigma x^2 = 1,836, \quad \Sigma y^2 = 587.05$$

With these sums and $n = 7$, the correlation coefficient is

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}}$$

$$\frac{7(435.6) - (88)(56.7)}{\sqrt{7(1,836) - (88)^2}\sqrt{7(587.05) - (56.7)^2}}$$

$r \approx -0.908$. this value indicates that there is strong negative relation between number of years out of school and annual contribution